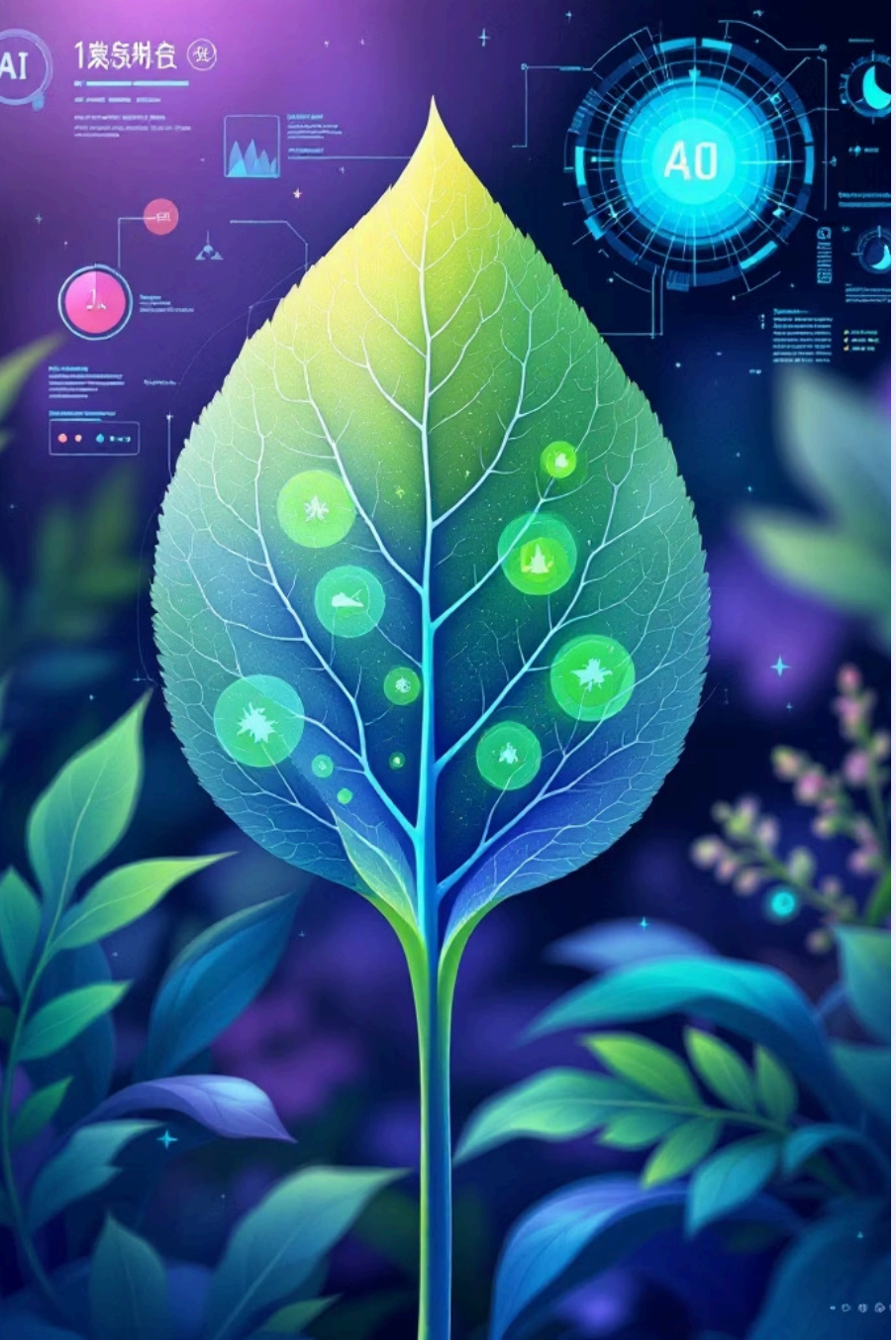


Plant Disease Captioning with BLIP-2 + LoRA

An end-to-end Vision Language Model pipeline for automated plant disease understanding using the PlantVillage dataset.

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Project Objectives



Shift from Classification to Captioning

Replace single-label disease prediction with rich, multi-faceted natural language descriptions of plant pathology



Semantic Understanding

Use Vision Language Models to understand disease context, not just surface-level visual patterns



Explainable AI

Generate human-readable captions that clearly explain the reasoning behind disease identification



Richer Feature Extraction

Capture subtle disease characteristics, severity, and progression that classification models often miss



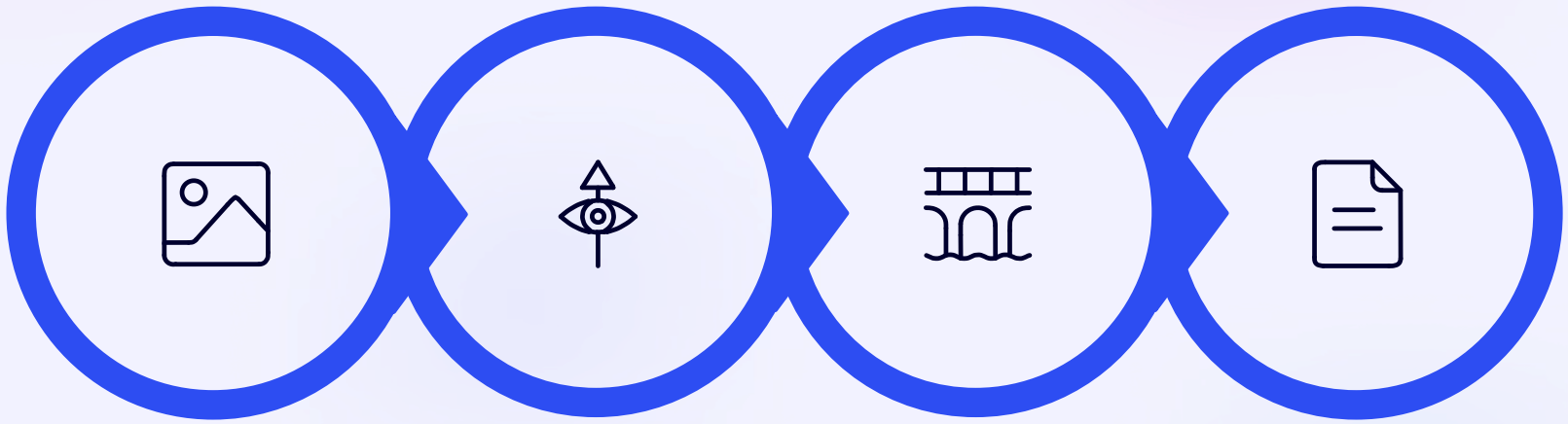
Complementary Approach

Use VLM captions as an alternative methodology to improve overall system robustness rather than an incremental classifier upgrade

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How Vision Language Models Generate Captions

Vision Language Models (VLMs) are pivotal for automated caption generation, combining visual and linguistic understanding. This pipeline details the process:



Each component plays a crucial role in generating accurate disease captions:

1) Image Input

This is where the plant leaf image enters the system, providing the visual data for analysis.

2) Vision Encoder (ViT)

The Vision Transformer (ViT) extracts sophisticated visual features from the input image, converting them into numerical representations.

3) Querying Transformer

This component acts as a bridge, translating the visual embeddings into query tokens that the language model can understand and process.

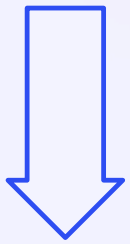
4) Language Model (OPT 2.7B)

Based on the query tokens, the large language model (OPT 2.7B) then generates the descriptive text captions for the identified plant diseases.

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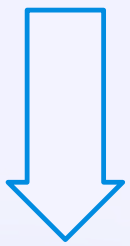
Methodology Pipeline Overview

Our approach integrates advanced vision language models with a structured data processing pipeline to accurately identify and caption plant diseases.



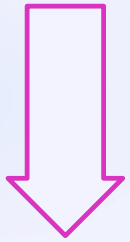
Data Input

Raw plant image data from the PlantVillage dataset is ingested.



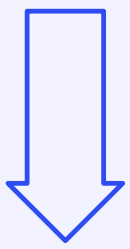
Data Augmentation

Images are augmented to increase dataset diversity and model robustness.



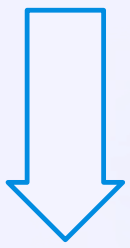
Sample Captions per Disease

A collection of representative captions is curated for each disease category.



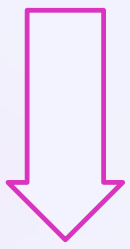
Caption Generation Script

Automated scripts generate synthetic captions based on augmented images and sample captions.



Fine-tuning

The BLIP-2 model is fine-tuned with LoRA using the generated captions and image data.



Inference

The fine-tuned model predicts disease captions for new, unseen plant images.

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Sample Disease Captions

Here are real example captions generated by our model for different plant diseases, illustrating the detailed descriptions provided.

Tomato Early Blight

Brown lesions with concentric rings on lower leaves, surrounded by yellow halo

Potato Late Blight

Water-soaked spots with white fuzzy mold on leaf undersides

Corn Common Rust

Cinnamon-brown pustules on both leaf surfaces

Apple Scab

Olive-green to brown velvety spots on leaves and fruit

Grape Powdery Mildew

White powdery patches on leaves and shoots

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Why BLIP-2 (2.7B) Model?

Our selection of the BLIP-2 2.7B model was driven by a combination of performance, efficiency, and adaptability, crucial for effective plant disease captioning.

Lightweight yet powerful

2.7B parameters is efficient for fine-tuning.

Vision-Language capability

Understands both images and text for comprehensive analysis.

Pre-trained on diverse data

Provides a strong foundation for transfer learning.

LoRA compatible

Enables parameter-efficient fine-tuning.

Fast inference

Suitable for real-time agricultural applications.



Future Enhancements for Better Accuracy

To further refine our plant disease captioning system and achieve even higher levels of accuracy, we plan to implement several key enhancements.

Larger Model Variants

Explore BLIP-2 7B or 13B for improved understanding and nuanced caption generation.

Multi-modal Fusion

Combine image data with metadata by manual caption feeding for richer context and more precise disease identification.

Ensemble Methods

Integrate multiple fine-tuned models to leverage their diverse strengths and boost overall robustness.

Advanced Augmentation

Utilize synthetic data generation and domain-specific augmentation techniques to expand dataset diversity.

Confidence Scoring

Incorporate uncertainty quantification to provide a measure of confidence with each generated caption.

Active Learning

Implement active learning strategies to prioritize hard examples for annotation, optimizing labeling efforts.

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Outcomes

Disease-Aware Captions

Structured descriptions identifying plant species, disease conditions, and severity levels

Classification via Captions

Text-based disease classification through generated descriptions

Evaluation-Ready Output

JSON results formatted for automated metrics and accuracy testing

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